

Programme	B. Sc. Computer Science				
Course Title	Digital Electronics and Computer Architecture				
Type of Course	Major				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Basic understanding of mathematical concepts, especially areas like algebra				
Course Summary	This course provides a comprehensive introduction to the fundamentals of digital systems, covering topics related to binary arithmetic, basic computer logic, combinational and sequential logic circuits, as well as basic computer organization and design. Throughout the course, students will gain a solid understanding of digital systems, from the basic building blocks of logic circuits to the design and organization of processors and memory				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand Basic Binary arithmetic Techniques	U	C	Instructor-created exams / Quiz
CO2	Implement logic operations using basic gates and Boolean algebra, design and optimise logic expressions using Karnaugh maps and design combinational logic circuits	Ap	P	Instructor-created exams/ Home Assignments
CO3	Understand the operation of latches and flip flops and the design of sequential logic circuits	U	C	Instructor-created exams

CO4	Learn the basic computer organization by understanding the role of registers, buses, ALU and control unit and the concepts like parallel processing and pipelining	U	C	Instructor-created exams
CO5	Understand how instructions represented, addressed and executed and how a microprogrammed control unit work	U	C	Instructor-created exams
CO6	Understand the concepts of memory and IO organization	U	C	Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Number systems and Boolean Algebra		10	15
	1	Binary arithmetic: Addition, Subtraction, Concepts of 1's and 2's complement, 1's and 2's complement addition	2	
	2	Logic Gates: AND, OR, NOT, NOR, NAND, XOR, XNOR , Universal Property of NAND and NOR gates	3	
	3	Boolean algebra: Boolean operations, laws and rules, Demorgan's theorem	2	
	4	Boolean Expression Simplification using K Map up to 4 variable	3	
II	Combinational and Sequential Logic Circuits		12	20
	5	Combinational Circuits: Half Adder, Full Adder, Ripple Carry Adder	1	
	6	Combinational Circuits: Encoder and Decoder (Basic Circuit Only)	1	
	7	Combinational Circuits: Multiplexer and Demultiplexer (Basic Circuit Only)	1	
	8	Concepts of Latches and Flip Flops, Types of Flip Flops (SR, D, JK, T): Truth Table and Circuit	3	
	9	Sequential Circuits: Synchronous and Asynchronous Counters	4	

	10	Johnsons and Ring counter, Shift Registers	2	
III	Basic Computer Organization and Micro Programmed Control		10	15
	11	Instruction codes, Registers and Common Bus system	2	
	12	Computer Instructions	1	
	13	Timing and Control: Concepts of hardwired and microprogrammed control	1	
	14	Instruction Cycle	1	
	15	Microprogrammed Control: Control memory & Address Sequencing	3	
	16	Micro Instruction Format and Symbolic Micro Instruction	2	
IV	Processor ,Memory and I/O Organization		16	20
	17	Processor Organisation: General Register organization and stack organization, Instruction formats and addressing modes	4	
	18	Processor Organisation: RISC vs CISC , Parallel Processing	2	
	19	Pipelining: General Considerations , Arithmetic Pipeline, Instruction Pipeline	3	
	20	Memory Organisation: Memory Hierarchy, Main Memory	1	
	21	Associative Memory, Cache Memory Mapping	4	
	22	IO Organisation: Modes of transfer: programmed IO, Interrupt initiated IO, DMA (Concepts Only)	2	
V	Open Ended Module: Computer Arithmetic & Types of Instruction		12	
	1	Computer Arithmetic: Addition and Subtraction, Multiplication Algorithms, Division Algorithms	7	
	2	Examples for Memory Reference, Register Reference, Input Output Instructions, Data Transfer Instructions, Data Manipulation Instructions , Arithmetic Instructions , Logical and Bit Manipulation Instructions , Shift Instructions, Program Control Instruction, Conditional Branch Instructions Subroutine Call and Return	5	

References

1. “Digital Fundamentals”, Thomas L. Floyd
2. “Computer System Architecture”, M. Morris Mano
3. “Computer Organization” , Carl Hamacher, Zvonko Vranesic

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	-	-	-	-	-						
CO 2	-	3	-	-	-	-						
CO 3	2	3	-	-	-	-						
CO 4	2	2	-	-	-	-						
CO 5	-	3	-	-	-	-						
CO 6	-	3	-	-	-	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓			✓